Netflix Business Analytics Project Report

In [42]:

```
#Import required libraries:
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns


#Ignore all incoming warnings (if any):
import warnings
warnings.filterwarnings("ignore")
```

In [43]:

```
#Load the dataset:
df = pd.read_csv("netflix_titles.csv")
```

Data inspection + Data preprocessing + Feature Engineering:

In [44]:

```
#First five rows:
df.head()
```

Out[44]:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	d
0	81145628	Movie	Norm of the North: King Sized Adventure	Richard Finn, Tim Maltby	Alan Marriott, Andrew Toth, Brian Dobson, Cole...	United States, India, South Korea, China	September 9, 2019	2019	TV-PG	90 min	Children & Family Movies, Comedies	
1	80117401	Movie	Jandino: Whatever it Takes	NaN	Jandino Asporaat	United Kingdom	September 9, 2016	2016	TV-MA	94 min	Stand-Up Comedy	r c
2	70234439	TV Show	Transformers Prime	NaN	Peter Cullen, Sumalee Montano, Frank Welker, J...	United States	September 8, 2018	2013	TV-Y7-FV	1 Season	Kids' TV	
3	80058654	TV Show	Transformers: Robots in Disguise	NaN	Will Friedle, Darren Criss, Constance Zimmer, ...	United States	September 8, 2018	2016	TV-Y7	1 Season	Kids' TV	p
4	80125979	Movie	#realityhigh	Fernando Lebrija	Nesta Cooper, Kate Walsh, John Michael Higgins...	United States	September 8, 2017	2017	TV-14	99 min	Comedies	r D

In [45]:

```
#Last 5 rows:
df.tail()
```

Out[45]:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	
6229	80000063	TV Show	Red vs. Blue	NaN	Burnie Burns, Jason Saldaña, Gustavo Sorola, G...	United States	NaN	2015	NR	13 Seasons	TV Action & Adventure, TV Comedies, TV Sci-Fi ...	Thi g
6230	70286564	TV Show	Maron	NaN	Marc Maron, Judd Hirsch, Josh Brener, Nora Zeh...	United States	NaN	2016	TV-MA	4 Seasons	TV Comedies	I st I i
6231	80116008	Movie	Little Baby Bum: Nursery Rhyme Friends	NaN	NaN	NaN	NaN	2016	NaN	60 min	Movies	ori fc
6232	70281022	TV Show	A Young Doctor's Notebook and Other Stories	NaN	Daniel Radcliffe, Jon Hamm, Adam Godley, Chris...	United Kingdom	NaN	2013	TV-MA	2 Seasons	British TV Shows, TV Comedies, TV Dramas	Se tl
6233	70153404	TV Show	Friends	NaN	Jennifer Aniston, Courteney Cox, Lisa Kudrow, ...	United States	NaN	2003	TV-14	10 Seasons	Classic & Cult TV, TV Comedies	Thi mis

In [46]:

```
#Detailed info:
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6234 entries, 0 to 6233
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       6234 non-null   int64 
 1   type          6234 non-null   object
 2   title         6234 non-null   object
 3   director      4265 non-null   object
 4   cast          5664 non-null   object
 5   country       5758 non-null   object
 6   date_added    6223 non-null   object
 7   release_year  6234 non-null   int64 
 8   rating        6224 non-null   object
 9   duration      6234 non-null   object
 10  listed_in     6234 non-null   object
 11  description   6234 non-null   object
dtypes: int64(2), object(10)
memory usage: 584.6+ KB
```

In [47]:

```
#Convert 'date_added" attribute to datetime type:
df['date_added'] = df['date_added'].astype(str).str.strip()
df["date_added"] = pd.to_datetime(df["date_added"])
```

In [48]:

```
#No. of missing values:
df.isnull().sum()
```

Out[48]:

```
show_id            0
type               0
title              0
director        1969
cast             570
country          476
date_added        11
release_year       0
rating            10
duration           0
listed_in          0
description        0
dtype: int64
```

In [49]:

```
#Data's size:
df.shape
```

Out[49]:

```
(6234, 12)
```

In [50]:

```
# Extract date features:
df['year_added'] = df['date_added'].dt.year
df['month_added'] = df['date_added'].dt.month
df['day_added'] = df['date_added'].dt.day

# Handle other missing values:
df['director'] = df['director'].fillna('Unknown')
df['cast'] = df['cast'].fillna('Unknown')
df['country'] = df['country'].fillna('Unknown')
df['rating'] = df['rating'].fillna('Not Rated')
df['description'] = df['description'].fillna('No description')

# Duration Feature Engineering:
df['duration_num'] = df['duration'].str.extract('(\d+)').astype(float)
df['duration_unit'] = df['duration'].str.extract('(min|Season)')

# Genre Feature Engineering (Important for Recommendation):
df['genres_list'] = df['listed_in'].str.split(', ')
df['num_genres'] = df['genres_list'].apply(lambda x: len(x) if isinstance(x, list) else
0)




def get_age_group(rating):
    if pd.isna(rating):
        return 'Unknown'
    mature = ['TV-MA', 'R', 'NC-17']
    teen = ['TV-14', 'PG-13']
    family = ['TV-PG', 'PG']
    kids = ['TV-Y', 'TV-Y7', 'G', 'TV-G']
    if rating in mature:
        return 'Mature'
    elif rating in teen:
        return 'Teen'
```

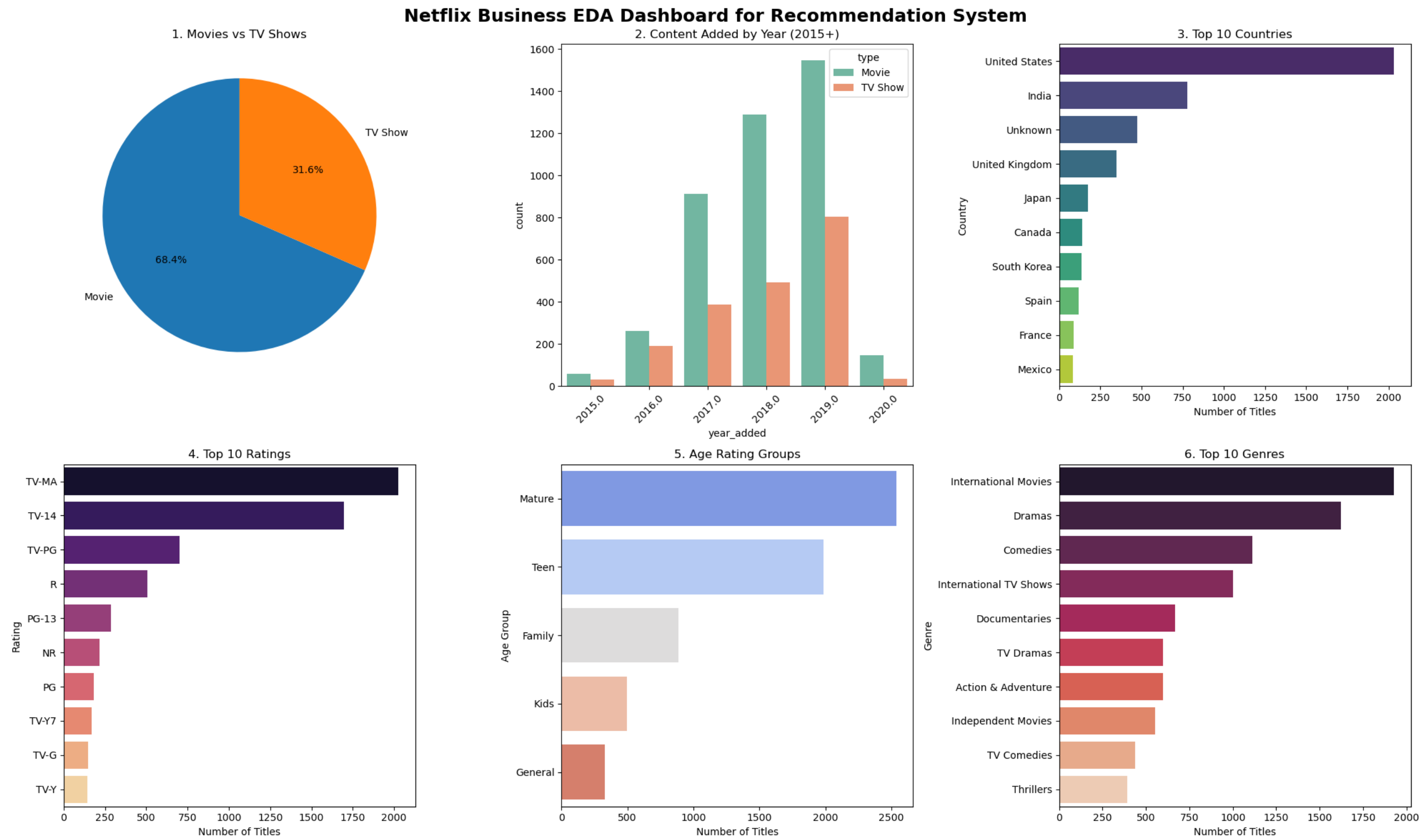
```python
    elif rating in family:
        return 'Family'
    elif rating in kids:
        return 'Kids'
    return 'General'

df['age_rating_group'] = df['rating'].apply(get_age_group)
```

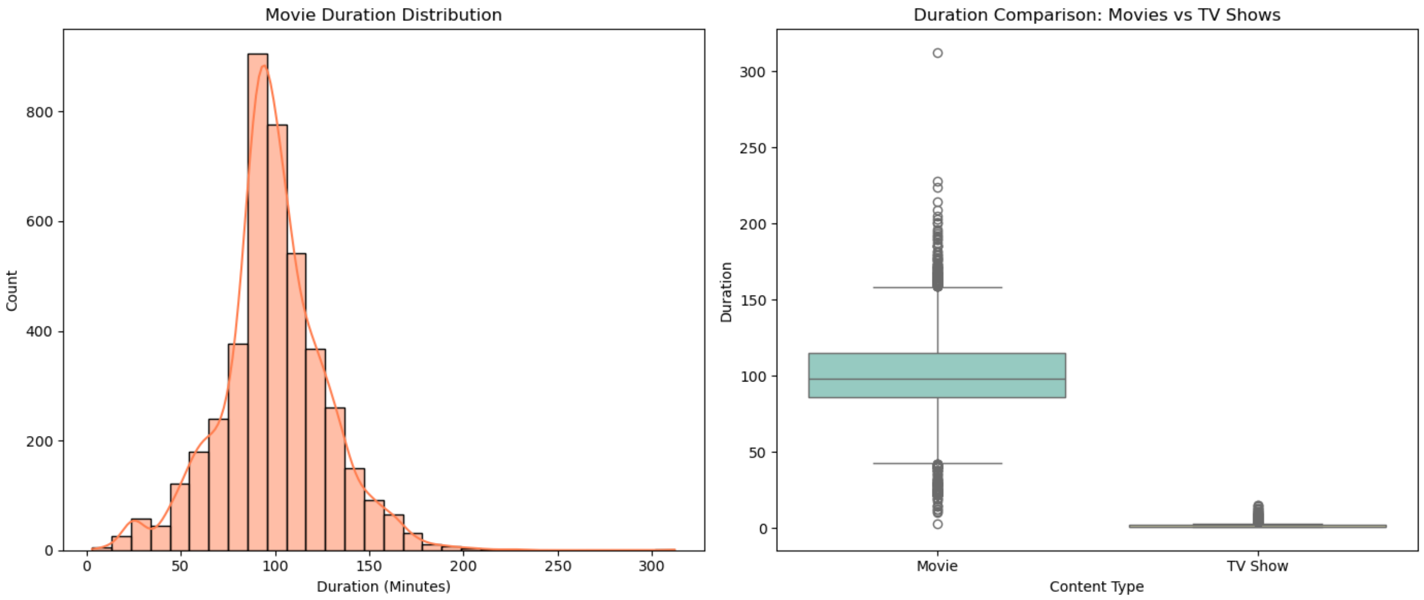
In [51]:

```python
plt.figure(figsize=(20,12))

plt.subplot(2,3,1)
content_type = df['type'].value_counts()

plt.pie(content_type.values,
        labels=content_type.index,
        autopct='%1.1f%%',
        startangle=90)

plt.title("1. Movies vs TV Shows")




plt.subplot(2,3,2)

recent = df[df['year_added'] >= 2015]

sns.countplot(data=recent,
              x='year_added',
              hue='type',
              palette='Set2')

plt.title("2. Content Added by Year (2015+)")
plt.xticks(rotation=45)




plt.subplot(2,3,3)

top_countries = df['country'].value_counts().head(10)

sns.barplot(x=top_countries.values,
            y=top_countries.index,
            palette='viridis')

plt.title("3. Top 10 Countries")
plt.xlabel("Number of Titles")
plt.ylabel("Country")




plt.subplot(2,3,4)

top_ratings = df['rating'].value_counts().head(10)

sns.barplot(x=top_ratings.values,
            y=top_ratings.index,
            palette='magma')

plt.title("4. Top 10 Ratings")
plt.xlabel("Number of Titles")
plt.ylabel("Rating")
```

```
plt.subplot(2,3,5)

age_group = df['age_rating_group'].value_counts()

sns.barplot(x=age_group.values,
            y=age_group.index,
            palette='coolwarm')

plt.title("5. Age Rating Groups")
plt.xlabel("Number of Titles")
plt.ylabel("Age Group")



plt.subplot(2,3,6)

genres_exploded = df.explode('genres_list')

top_genres = genres_exploded['genres_list'].value_counts().head(10)

sns.barplot(x=top_genres.values,
            y=top_genres.index,
            palette='rocket')

plt.title("6. Top 10 Genres")
plt.xlabel("Number of Titles")
plt.ylabel("Genre")

plt.suptitle("Netflix Business EDA Dashboard for Recommendation System",
             fontsize=18,
             fontweight='bold')

plt.tight_layout()

plt.savefig("netflix_business_eda_dashboard.png",
            dpi=300,
            bbox_inches="tight")

plt.show()



plt.figure(figsize=(14,6))
plt.subplot(1,2,1)

sns.histplot(
    df[df['type']=="Movie"]['duration_num'].dropna(),
    bins=30,
    kde=True,
    color='coral'
)

plt.title("Movie Duration Distribution")
plt.xlabel("Duration (Minutes)")
plt.ylabel("Count")

plt.subplot(1,2,2)

sns.boxplot(
    data=df,
    x='type',
    y='duration_num',
    palette='Set3'
)

plt.title("Duration Comparison: Movies vs TV Shows")
plt.xlabel("Content Type")
plt.ylabel("Duration")

plt.tight_layout()
```

```
plt.show()
```





In [ ]:

Business Report:

- **Movies are the dominant category of content.**
- **From 2017 - 2019 we observed that a lot of new content was added to the content library.**
- **Most titles belong to the United States.**
- **TV-MA and TV-14 is dominant rating types.**
- **Most titles belong to eitheir mature or teen category.**
- **International Movies , Dramas and Comedies are the dominant genres.**
- **Most movies have a runtime between 80 and 120 minutes, indicating a preference for standard feature-length content.**
- **Movies generally have significantly longer durations than TV Shows, while TV Shows are typically limited to a small number of seasons, with only a few long-running series appearing as outliers.**

In [ ]:

```

```

Project by:

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